

Mood-Based Playlist Generator

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Background

In today's digital age, personalization has become a fundamental aspect of user experience across various platforms. From personalized advertisements on social media to recommended movies on streaming services like Netflix, data-driven customization plays a vital role in enhancing user satisfaction.

Machine learning algorithms are key to creating personalized experiences. They help systems learn from users' actions and preferences, allowing them to make better predictions. By looking at large amounts of data on user behavior, machine learning models can find patterns that help make recommendations. Various machine learning models can help create recommendation systems that predict what products, movies, or songs a user might like based on their past behavior or what similar users enjoy. These models get better over time as they process more data, ensuring the recommendations stay relevant as users' tastes change. In music, machine learning can be used to predict and classify moods or genres from audio features, lyrics, and even the user's mood, making playlist generators more personalized and engaging [1][2].

The Project

Our project builds on this idea by combining sentiment analysis, TF-IDF, and Random Forest models to create a playlist generator. By understanding both the mood of songs and the emotions of users, we aim to create playlists that feel more personal and emotionally connected [1].

Sentiment analysis will classify an input text from the user describing their mood to help identify the emotional tone. Random Forest models will be used to classify songs based on selected musical

features. In addition, TF-IDF will be applied to the song lyrics to capture more of the emotional content [2]. By combining these methods, the system will better match songs to the user's described mood.

Research question

- Can the machine learning models Random Forest, TF-IDF and sentiment analysis accurately classify song moods and generate personalized playlists?
- What are the limitations and opportunities of combining sentiment analysis, TF-IDF and Random Forest to understand and predict users' emotional states for music recommendations?

Dataset

The dataset used in this project is a collection of songs, each with a variety of features that provide insights into both the audio properties and lyrical content of the music. It combines both numerical and textual data to describe various aspects of each track. Key columns in the dataset that is included in this project:

Artist Name and Track Name	Identifies the song and its creator.
Lyrics	Provides the song’s lyrical content, which we analyze using text-based techniques like TF-IDF.
Valence	Measures how positive or happy the song feels.
Energy	Captures the intensity or activity level of the song.

Figure 1. Key columns in the dataset

Preprocessing

The process begins by loading the dataset and retaining only the most relevant features, such as numerical audio properties and textual data and irrelevant columns are removed. The features, shown in Figure 1, play a crucial role in analyzing the mood of songs and matching them to user emotions.

Method

To classify the mood of songs in the dataset, audio features were combined with text-based data. Audio features like valence and energy were essential in identifying the emotional tone and intensity of each song. A rule-based system was designed to assign moods based on specific thresholds:

	Valence	Energy
Happy	>0.7	>0.6
Sad	<0.4	<0.5
Angry	<0.3	>0.7
Calm	-	-

Figure 2. Thresholds in rule-based system

Sentiment Analysis

Sentiment analysis is a method used to understand the emotional tone of a text and classify it as positive, negative, or neutral. In this project, we used VADER (Valence Aware Dictionary and Sentiment Reasoner) from the NLTK library to analyze the user’s mood description. VADER assigns a sentiment score to the text, which helps determine whether the mood is "Happy," "Angry," "Sad," or "Calm." The score is a number that reflects how positive, negative, or neutral the input is, allowing the system to understand the user's mood even if they do not use specific keywords. VADER is particularly effective for understanding the sentiment of short, text-based inputs like user mood descriptions. The sentiment analysis step is crucial because it becomes the foundation for generating a playlist that is tailored to their emotional state and personalization in our recommendations.

TF-IDF & Random Forest

To classify the mood of songs in our dataset, we used a combination of text and numerical features. For text-based features, we applied the TF-IDF method. This technique converts song lyrics into numerical values by measuring the importance of each word within the dataset. By using TF-IDF, we enable our model to interpret the lyrical content of songs in a good way. In addition to TF-IDF, we used audio features like **valence**, and **energy**, which provide numerical insights into the characteristics of each song. These features, combined with the lyrical data, were fed into a Random Forest classifier. The classifier was trained using a portion of the dataset and tested on a separate portion to ensure its accuracy. Through this process, the model learned to label songs based on their mood, combining the textual and numerical features into a comprehensive mood classification system.

Result

The project successfully implemented a playlist generator that use machine learning techniques to classify song moods and align them with user emotions. Using the Random Forest model combined with TF-IDF and sentiment analysis, the system achieved high accuracy in classifying moods such as "Happy," "Sad," and "Calm," with a classification accuracy of 99%.

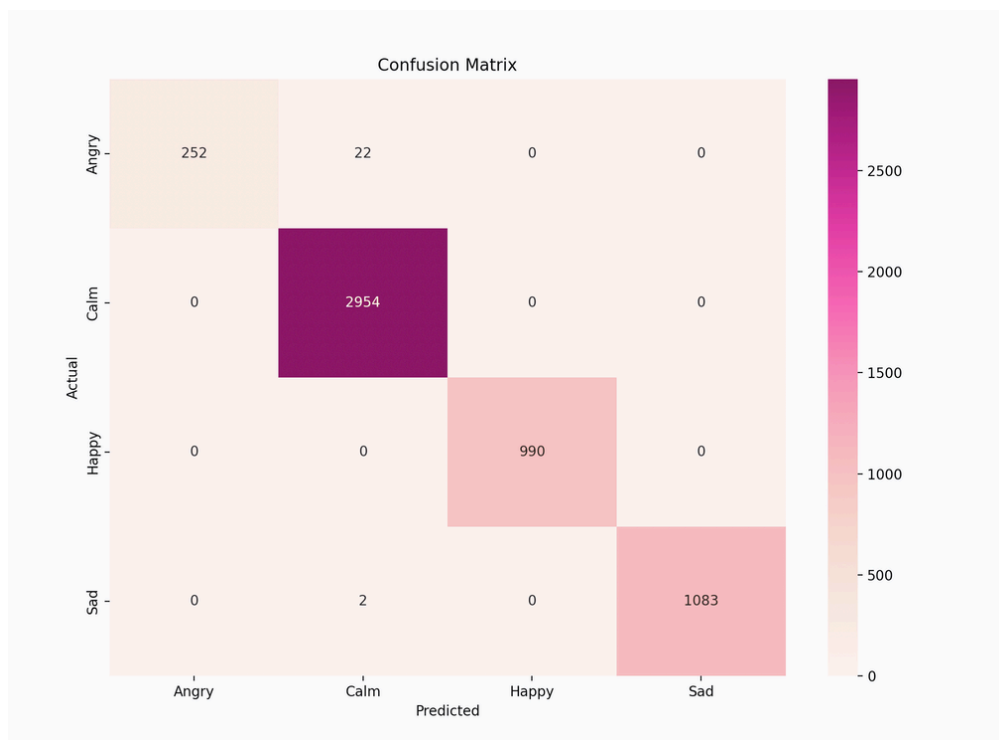


Figure 3. Confusion matrix

The confusion matrix and classification report confirmed the model's effectiveness, particularly for common moods, although there were slight challenges in accurately classifying less frequent moods like "Angry."

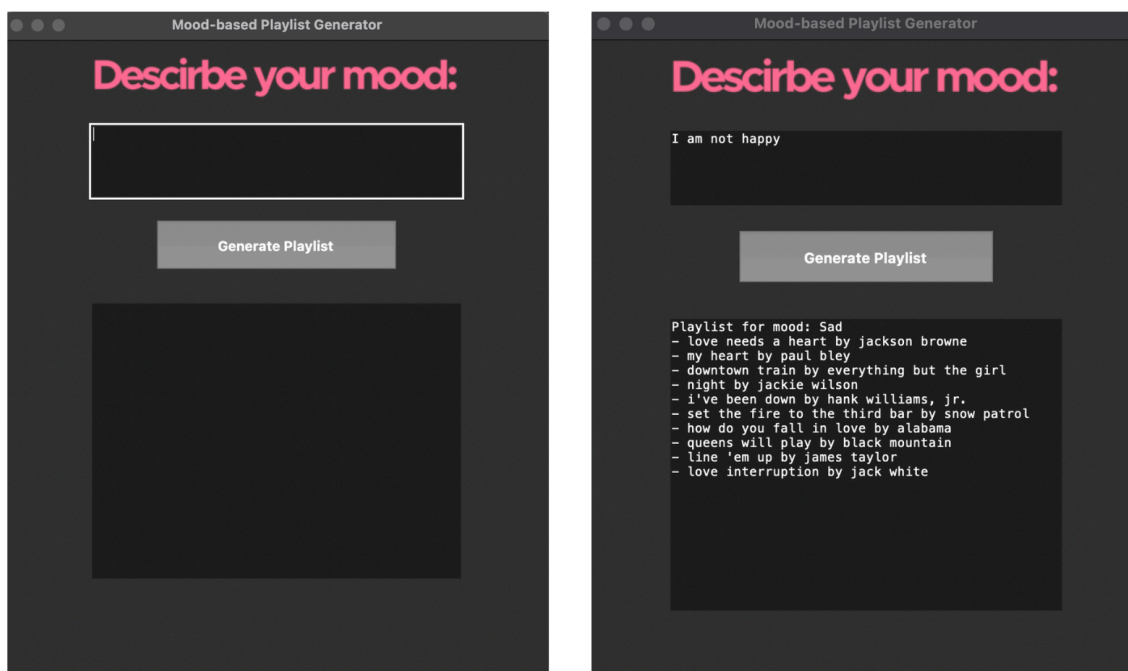


Figure 4. Mood-Based Playlist Generator User Interface

The GUI allows users to describe their emotional state in text, which is analyzed using the VADER sentiment analyzer. This analysis maps the input to a mood, such as "Sad" or "Happy," and generates a playlist tailored to the user's emotional tone. For instance, a user input of "I am not happy" successfully generated a playlist of songs classified under the "Sad" mood.

Discussion

Random Forest, TF-IDF, and sentiment analysis successfully classified song moods and created personalized playlists. The algorithm performed well in identifying the mood in both user input and songs. However, there are still areas for improvement that need to be addressed.

Subjectivity issue

One of the key challenges in this project is the subjectivity of music and mood perception. Music is a personal experience, and different individuals may have varying emotional responses to the same song. While our system classifies songs into moods such as "Happy," "Sad," or "Angry" by analyzing a combination of audio features, lyrics, and machine learning models, the emotional labels assigned to each song are not universally agreed upon. The way one person interprets a song's mood may differ significantly from another's interpretation. For example, a song that one listener associates with feelings of happiness might evoke sadness or nostalgia in someone else, depending on their personal experiences, cultural background, or context [3]. This subjectivity makes mood classification a complex task, as it is influenced by not only the objective features of the song but also individual emotions, memories, and associations.

Improvements with personalization

The algorithm has areas of improvement regarding the accuracy and personalization. The algorithm can be made more personalized to each user by increasing interesting features to match the user's mood description. Including features like **tempo**, **key**, and **liveness** could capture a broader range of musical properties. Currently, the only possible mood categories are four, very general ones. Increasing these categories to more specific ones, can ultimately increase the chances of the mood classification to hit right.

Integrating a user's previous listening behavior could also make the agent more personalized to each user. Research shows that familiar music has a strong effect on emotions by activating parts of the brain linked to feelings and memory [4]. Familiar songs often create stronger emotional responses than unfamiliar ones because they are easier to process and connect with past experiences [4]. This suggests that including familiar songs in mood-based playlists could make them more effective and enjoyable for users.

The agent could be improved by giving the user the choice of getting a playlist that matches the mood or one that can counteract the mood. If a user describes themselves as sad, they might not want to listen to sad music. They might instead have a desire to listen to uplifting and happy music to feel better.

Conclusion

This project successfully combined Random Forest, TF-IDF, and sentiment analysis to classify song moods and generate mood-based playlists. The models achieved high accuracy in predicting moods like "Happy" and "Sad," showing the effectiveness of using both audio features and lyrics. However, limitations include the subjectivity of mood interpretation and difficulties in classifying less common moods like "Angry." Future work could include additional features, such as user preferences or song genres, to enhance accuracy and personalization.

Research Questions

- Can the machine learning models Random Forest, TF-IDF and sentiment analysis accurately classify song moods and generate personalized playlists?

The combination of Random Forest, TF-IDF, and sentiment analysis has been successful in classifying song moods and creating personalized playlists. By looking at both the audio features and the lyrics of songs, the models were able to predict moods like "Happy" and "Calm" with high accuracy. Sentiment analysis helped understand the emotional tone of user input which improved the system's ability to match songs to the right mood. Overall, the approach worked well for classifying moods and generating playlists based on those moods.

- What are the limitations and opportunities of combining sentiment analysis, TF-IDF and Random Forest to understand and predict users' emotional states for music recommendations?

Although the combination of sentiment analysis, TF-IDF, and Random Forest has shown good results, there are some limitations. One challenge is that people perceive music differently and therefore, moods may be interpreted in various ways by different listeners. Also, some moods, like "Angry," are harder to predict because it does not appear as often in the data. However, there are opportunities for improvement. Future work could include adding user preferences or specific genres to make the mood classification more accurate and personalized. This would help the system recommend songs that better match each user's individual taste and emotional state.

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